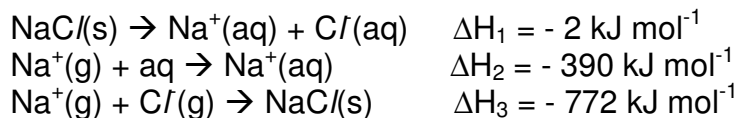


Answer any **four** questions.

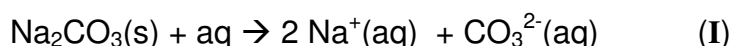
- 1 (a) Given the following reactions



- (i) Name ΔH_1 and ΔH_3 .
- (ii) Draw an appropriate energy cycle to calculate the enthalpy change of hydration of the chloride ion.

[4]

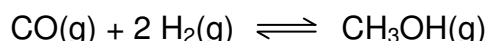
- (b) When 2.12 g (0.02 mol) of sodium carbonate was added to 30.0 cm³ of water, the temperature fell by 5.2 °C.



- (i) Calculate the enthalpy change of this reaction per mol of Na₂CO₃. Assume that the specific heat capacities of all solutions are 4.2 J g⁻¹ K⁻¹ and that all solutions have a density of 1.0 g cm⁻³.
- (ii) Explain why ΔS of reaction (I) is + 154.3 J K⁻¹ mol⁻¹ and hence calculate the minimum temperature for reaction (I) to be spontaneous.

[4]

- (c) CO is used in the manufacture of methanol according to the following reaction:



A 40.8 mol sample of CO was added to 146 mol of H₂. When equilibrium is reached at 400 °C, the mixture contained 30.2 mol methanol and the total pressure was 7.50 MPa.

- (i) Calculate the equilibrium constant K_p for this reaction.
- (ii) In order to increase the yield of methanol, H₂ is added to the equilibrium system. Predict, with reasons, whether this attempt is valid.

[3]

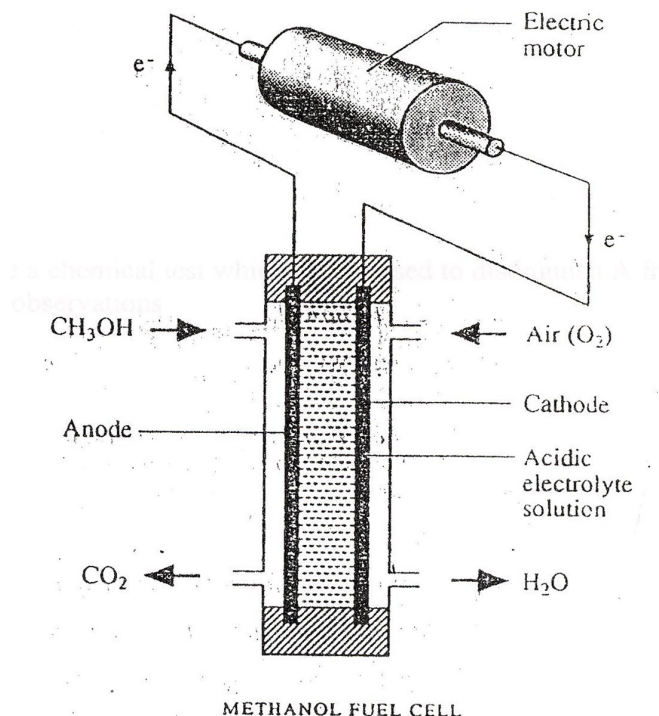
- (d) CH₃OH has been found to be useful in the synthesis of other organic compounds. In one process, CH₃OH is used as a raw material for the preparation of organic solvent dimethyl ether, CH₃OCH₃.

Suggest a suitable synthetic route leading to the formation of CH₃OCH₃.

[3]

- (e) Scientists are hoping to use methanol to power a fuel cell in the car instead of burning petrol in an internal combustion engine. The fuel cell generates an electric current which drives an electric motor.

One possible design of a methanol fuel cell under investigation is shown below.



The fuel and oxygen are fed continuously to the two electrodes. The electrodes are made of platinum dispersed onto a porous carbon support.

- (i) Write half equations for the reaction which takes place at the electrodes of the fuel cell and hence an overall equation for the cell reaction.
- (ii) Suggest one reason why methanol is a better choice than petrol as a fuel for motor cars.
- (iii) One refinement of fuel cell design is to replace the acidic electrolyte solution with a film of solid hydrogen ion conducting polymer. Explain why this would be an improvement.
- (iv) One fuel cell in use now is the hydrogen-oxygen fuel cell. Give an advantage of this over conventional batteries and cells.

[6]

[Total: 20]

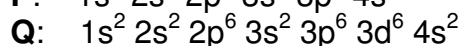
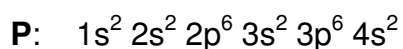
- 2 (a) The boiling points of the elements in the second period are shown in the table below.

Element	Li	Be	B	C	N	O	F	Ne
bp/ $^{\circ}\text{C}$	1330	2477	3930	4830	-196	-183	-188	-246

- (i) Explain the variation in the boiling points in terms of their structure and bonding of elements.
- (ii) Give the equations to show the reactions of Li_2O and CO_2 with water.

[5]

- (b) Consider the two metals whose electron configurations are as follows:



- (i) Compare the atomic radii of the two elements.
- (ii) State and explain in terms of electronic structure, one characteristic property of **Q** (other than that in b(i)) which is markedly different from those of **P**.

[3]

- (c) C and Si belong to the same group and yet the oxide of C exists as a gas while that of Si exists as a solid at room temperature. Suggest a reason for their physical states.

[2]

- (d) In an attempt to extract Li from its ore, Li_2O , a student suggested dissolving Li_2O in water, followed by electrolysis of the aqueous solution he had prepared.

- (i) With the aid of reduction potentials from the Data Booklet, explain whether he would be successful in his attempt.
- (ii) At the anode, a gas evolved was measured at s.t.p. Calculate the volume of this gas obtained if a current of 2 A was passed through the solution for a duration of 1 hour.

[4]

- (e) Solid barium amide, $\text{Ba}(\text{NH}_2)_2$ decomposes when heated to give barium nitride Ba_3N_2 and $\text{NH}_3(\text{g})$.
- (i) Draw the dot cross diagram for Ba_3N_2 .
- (ii) Write a balanced equation for the decomposition of barium amide.
- (iii) Based on your knowledge of the thermal decomposition of group II nitrates and by reference to the equation in e(ii), explain why the decomposition is favourable.
- (iv) What do you think will be the decomposition temperature of LiNH_2 when compared to $\text{Mg}(\text{NH}_2)_2$? Explain your answer.

[6]

[Total: 20]

- 3 (a) Describe what happens when concentrated sulphuric acid is added to solid sodium salts of chloride, bromide and iodide respectively. Give relevant equations.

[3]

- (b) Under suitable conditions, iodine and chlorine react to give an interhalogen compound, ICl_x . When a pure sample of ICl_x was dissolved in an excess of aqueous potassium iodide, the iodine liberated required 17.1 cm^3 of 0.1 mol dm^{-3} sodium thiosulphate solution for titration whereas the chloride ion liberated required 12.8 cm^3 of 0.1 mol dm^{-3} aqueous silver nitrate for complete precipitation.

Determine the value of x . Hence, write a balanced equation for the reaction between ICl_x and I^- .

[4]

- (c) Silver chloride has been found to be a sparingly soluble salt. Given the solubility product of silver chloride is $1.6 \times 10^{-5} \text{ mol}^2 \text{ dm}^{-6}$ at a given temperature.

(i) Calculate the solubility of silver chloride in mol dm^{-3} in water at this temperature.

(ii) Calculate the minimum concentration of dilute HCl(aq) needed to precipitate out silver chloride if 2 cm^3 of the acid is to be added to 10 cm^3 of a solution containing 0.1 mol dm^{-3} of silver nitrate.

(iii) When concentrated hydrochloric acid is gradually added to a solution of silver nitrate, a white precipitate is first formed but this dissolves when aqueous ammonia is added in excess. Explain this observation.

[5]

- (d) Given the following third ionisation energies for the elements Ca, Mn and Fe.

Element	Third IE / kJ mol^{-1}
Ca	4940
Mn	3250
Fe	2960

Comment on the differences in the third ionisation energies for the following pairs.

(i) Ca and Mn

(ii) Mn and Fe

[3]

(e)

Ligand	NH ₃	EDTA ⁴⁻
log K (K = stability constant)	2.7	18.6

A solution contains 0.100 mol of aqueous NH₃ and 0.100 mol of EDTA⁴⁻ dissolved in water. 0.01 mol of Ni²⁺ was then added.

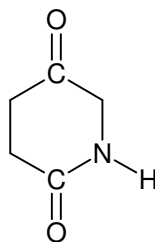
What would you expect to be the main complex ion in the solution? Give a reason for your answer.

[2]

(f) Give a reason why copper is a transition element.

[1]

(g)



Give the structure of the organic product formed when the above compound reacts with

(i) Hot aqueous NaOH

(ii) 2,4 - DNPH

[2]

[Total: 20]

- 4 (a) The lattice energies of the potassium halides are given in the table below.

Compound	Lattice energy / kJ mol^{-1}
KF	-801
KBr	-670
KI	-629

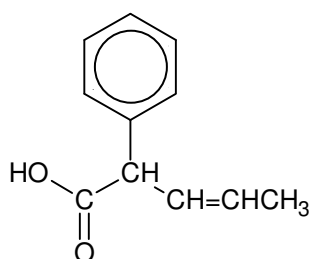
Suggest an explanation for the trend shown in the lattice energies.

[2]

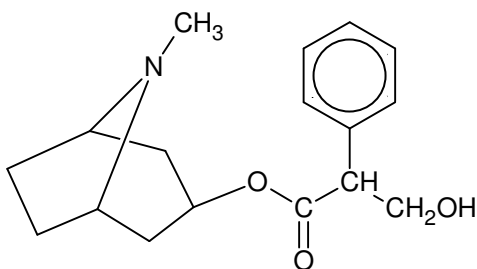
- (b) Nitrogen reacts with chlorine to form NCl_3 but phosphorus can react with chlorine to form both PCl_3 and PCl_5 . Explain the difference in their behaviour.

[2]

- (c) A compound **Y**

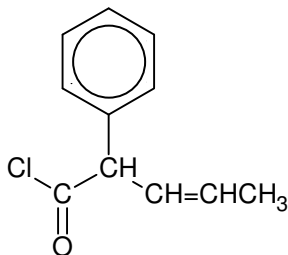


can be derived from atropine



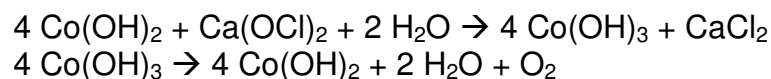
Atropine is a very toxic drug but dilute solutions of it is used during eye examination to dilate the pupil of the eye.

- (i) State the types of stereoisomerism **Y** displays. Draw diagrams to represent them.
- (ii) Describe reagents and conditions used to produce the following compound from **Y**.



[6]

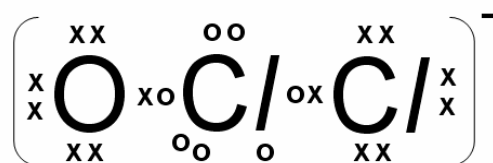
- (d) The decomposition of bleaching powder (essentially calcium chlorate (I)) occurs by the following steps:



- (i) Using **one** appropriate **diagram**, explain how Co(OH)_2 serves a specific role in the reaction shown.

[5]

The dot-cross diagram of an anion $\text{OC}l_2^-$ derived from OCI^- is shown below:



This anion is found to react with methane spontaneously. A sample of methane is reacted with Cl_2 , contaminated with $\text{OC}l_2^-$, in the presence of light.

- (ii) With the aid of equations, give the mechanism in the reaction of Cl_2 with methane.

[3]

- (iii) In reaction (ii), a small quantity of $\text{CH}_3\text{Cl}_2\text{O}^-$ is formed. Account for this observation with an appropriate equation.

[2]

[Total: 20]

- 5 (a) (i) What is meant by the primary structure of a protein?
- (ii) The folding of the α – helix structure in proteins such as enzymes is caused by interactions of the side-chain (R) groups. Some situations however can destroy the R group interactions. Identify the type of R group interaction that is destroyed in each of the following situation.
- (A) Heating
 (B) High acidity
 (C) Presence of cations such as Hg^{2+} or Cu^{2+}
- (iii) 1 mole of a penta-peptide gave the following products on complete hydrolysis: 2 moles of glycine (G), 1 mole of alanine (A), 1 mole of serine (S) and 1 mole of leucine (L). On partial hydrolysis, dipeptides alanylglycine (A-G) and leucylglycine (L-G) were obtained. Partial hydrolysis under another condition gives a dipeptide serylleucine (S-L) and the tripeptide glycylserylleucine (G-S-L).

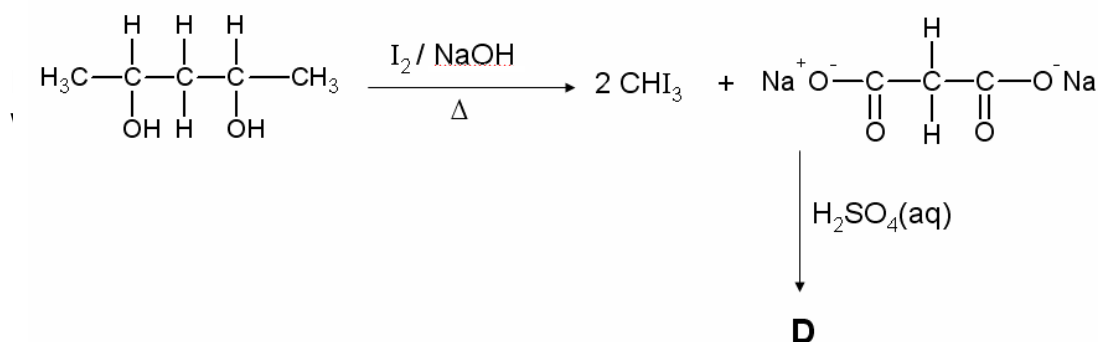
From these results, determine the sequence of amino acids in the penta-peptide.

[5]

- (b) An organic compound **A**, $\text{C}_5\text{H}_6\text{O}_4$, does not react with aqueous sodium carbonate but reacts with hot aqueous NaOH to form **B**, $\text{Na}_2\text{C}_3\text{H}_2\text{O}_4$, and **C**, $\text{C}_2\text{H}_6\text{O}_2$, which on boiling with acidified potassium dichromate (VI) gave ethanedioic acid, $(\text{COOH})_2$.

B on acidification gave **D**, $\text{C}_3\text{H}_4\text{O}_4$, which reacts easily with aqueous sodium carbonate. 1 mole of **D** gave white fumes of HCl and **E**, $\text{C}_3\text{H}_2\text{O}_2\text{Cl}_2$, when treated with 2 moles of PCl_5 .

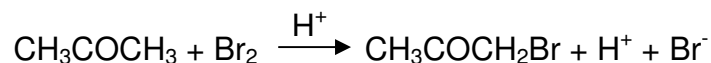
D can also be synthesised by the following reaction pathway:



Identify compounds **A** to **E**, explaining your reasoning and showing equations where appropriate.

[7]

(c) The bromination of acetone is acid-catalysed:



The rate of disappearance of bromine was measured for several different concentrations of acetone, bromine and hydrogen ions at a certain temperature.

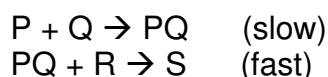
Expt No.	$[\text{CH}_3\text{COCH}_3]$ /mol dm ⁻³	$[\text{Br}_2]$ /mol dm ⁻³	$[\text{H}^+]$ /mol dm ⁻³	Rate / mol dm ⁻³ s ⁻¹
1	0.30	0.05	0.05	5.7×10^{-5}
2	0.30	0.10	0.05	5.7×10^{-5}
3	0.30	0.05	0.10	1.2×10^{-4}
4	0.40	0.05	0.20	3.1×10^{-4}
5	0.40	0.05	0.05	7.6×10^{-5}

- (i) Determine the order of reaction with respect to each reactant and hence the rate equation.
- (ii) In another experiment, the concentration of CH_3COCH_3 used was in excess. As a result, a new rate law was established as follows:

$$\text{Rate} = k' [\text{H}^+]$$

If the initial concentration of H^+ ions is 1.6 mol dm^{-3} and the half-life for the reaction is 5 minutes, sketch a graph of concentration of H^+ against time for the first 20 minutes of the reaction

- (iii) Assigning H^+ as **P** and CH_3COCH_3 as **Q**, the following elementary steps in the mechanism are as shown:



Given that the activation energy of the reaction is $+35 \text{ kJ mol}^{-1}$ and ΔH is -90 kJ mol^{-1} , sketch the potential energy profile diagram for the above reaction.

[8]

[Total: 20]

- End of Paper -